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by [Ali Alzahmi](#) - Wednesday, 6 August 2025, 8:58 PM

Modern computational problems have become highly decentralised, dynamic, and complex, thus making traditional monolithic and centralised systems obsolete. The ineffectiveness of this was what spawned Agent-Based Systems (ABS) as a way of addressing this shortcoming, or rather, fulfilling the demands of autonomous, contextually aware, socially capable software entities that can explore distributed environments, negotiate with peers, and learn through interactions (Gerges et al., 2021). In contrast with rule-based or procedural models, the agents are functional in autonomy, proactivity, reactivity, and social competence. These features also enable the agents to be agentic and act without being continuously supervised, and to reason in a world of uncertainty and change. Consequently, ABS can be used well to model systems with heterogeneous actors, conflicting objectives, and weak global depth, e.g., smart grids, financial markets, and disaster response coordination (Pulikottil et al., 2023).

The explosion in connectivity and processing power has also spurred this transition. Along with the advent of the Internet of Things (IoT), edge computing, and multi-agent simulations, the agents have become smart sensors in social-technical ecosystems. To mention a few, urban mobility agents can model individual vehicles optimising their routes dynamically in the context of current traffic in the city, thereby smoothly optimising the flow without involving any centralised control (Hayes & Grüter, 2023). The organisational advantages are not one-dimensional. ABS has the scalability of implementing agents on separate modules, resilience of having distributed decision-making, and offers adaptability due to the continuous learning and feedback (Yao et al., 2023). They allow bottom-up evolution of systems, local dynamics give rise to global phenomena, which facilitates more sensitive, responsive, and stable operation.

However, unlike fixed automation pipelines, agent-based computing provides organisations with a paradigm of intelligent autonomy so that systems become capable of operating with a level of foresight, negotiation, and environmental sensitivity that used to be lacking (Ocker et al., 2021). ABSs are not just devices; architecture changes toward more realistic and context-savvy computational ecologies.

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